NISHITA RANA

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EDUCATION

National Institute of Technology, Andhra Pradesh (2026)
B.Tech in Electronics and Communication Engineering (CGPA: 7.33)
Sacred Heart Convent School, Chandausi Moradabad, UP
ICSE Class 12: 93.75%
ICSE Class 10: 92.8%

WORK EXPERIENCE

IISER Bhopal | *Project: Human Action Recognition using CNN and LSTM* **May'25 - July'25**

- Built a deep learning–based Human Activity Recognition system using a **ConvLSTM2D** architecture to capture spatial and temporal features from video sequences.
- Preprocessed video data by extracting fixed-length 20-frame sequences, resizing to 64×64, and normalizing pixel values.
- Designed a 3-layer ConvLSTM2D model (64–128–256 filters) followed by MaxPooling3D and Dense softmax classifier for multi-class prediction.
- Implemented Early Stopping (patience=15) and trained the model using the Adam optimizer with categorical cross-entropy loss.
- Achieved ~77% test accuracy with a 75:25 train-test split and evaluated performance through validation loss analysis.

ACADEMIC PROJECTS

IoT-Based Heart Rate Monitoring System **Jan'25 - Apr'25**

- Developed an IoT-based heart rate monitoring system with **ESP8266** and a **DFRobot heart rate sensor** to capture real-time photoplethysmography (PPG) signals and compute BPM values.
- Designed embedded firmware in Arduino to acquire, process, and transmit pulse data for further analysis.
- Preprocessed and analyzed raw PPG signals in Python, applying feature extraction and Principal Component Analysis (PCA) for dimensionality reduction.
- Built and evaluated a predictive model on processed features, achieving a Brier score of ~0.03, indicating strong probabilistic calibration.

Healthcare Chatbot **Jan'24 - Apr'24**

- Developed an intent-based healthcare chatbot leveraging **NLP** techniques to classify user queries and generate appropriate responses.
- Utilized a structured intent dataset with multiple tagged patterns and responses to enable multi-class intent classification.
- Applied text preprocessing techniques including **tokenization**, **stemming**, bag-of-words vectorization, and one-hot encoding of intent labels.
- Trained a feedforward neural network using Keras for multi-class intent classification and integrated dynamic response generation.

SKILLS

- **Programming Languages:** C++, Python
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems
- **Machine Learning:** Supervised Learning, Classification, Neural Networks, CNNs, Natural Language Processing (NLP)
- **AI Frameworks & Tools:** TensorFlow, Keras, NumPy, Pandas

ACHIEVEMENTS

- Maintained consistent academic performance with no backlogs.
- Executive Member of the Artificial Intelligence and Robotics Club, NIT Andhra Pradesh.
- Completed NPTEL courses in DBMS and Introduction to Operating Systems.